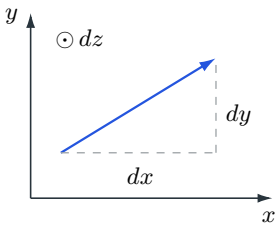
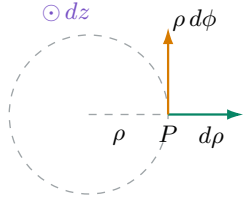
电场线方程使用实际线元：角度增量要乘相应长度

直角坐标



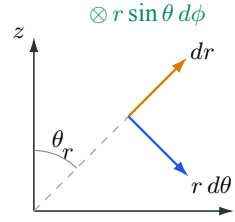
$$\frac{dx}{E_x} = \frac{dy}{E_y} = \frac{dz}{E_z}$$

圆柱坐标（俯视）



$$\frac{d\rho}{E_\rho} = \frac{\rho\, d\phi}{E_\phi} = \frac{dz}{E_z}$$

球坐标（子午面）



$$\frac{dr}{E_r} = \frac{r\, d\theta}{E_\theta} = \frac{r\sin\theta\, d\phi}{E_\phi}$$

各小图显示局部正方向；分母为零时回到参数方程判断，不能直接相除。